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AshGdNutDefAM: Machine-learning-based multi-feature framework for nutrient deficiency detection and classification in Ash gourd Plant

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ABSTRACT

The sustainability of agricultural production is increasingly threatened by nutrient deficiencies that compromise crop health, quality, and yield. Optimal crop management requires early detection and classification of these nutrient deficiencies. Traditional methods employ laboratory testing to predict nutrient deficiency in plant samples. Computer vision models provide an alternative to visual inspection by analysing and predicting the presence of nutrient deficiencies. A hybrid feature-based learning framework, AshGdNutDefAM, is proposed for the automated detection of nutrient deficiencies in the Ash gourd (*Benincasa hispida*) plant variety as part of the study. A rule-based and vein-based feature extraction approach is proposed, encompassing domain-specific botanical knowledge along with conventional features such as colour, texture, and spatial descriptors to provide an exhaustive representation of nutrient deficiency symptoms. Key physiological characteristics, such as the green-to-yellow pixel ratio, extent of interveinal chlorosis, and the dimensions of necrotic margins, that are associated with macronutrient and micronutrient deficiencies, especially magnesium (Mg) and iron (Fe), are captured by rule-based features. The proposed framework extracts 55 discriminative features across five groups: colour, spatial, texture, vein, and rule-based features. These features are evaluated using three machine learning classifiers: Support Vector Machine (SVM), Random Forest, and Gradient Boosting. The dataset collected includes healthy, magnesium and iron deficiency of the Ash gourd plant variety captured from agricultural farms in Mysuru, India. The performance of the proposed feature set is further compared with conventional handcrafted feature extraction techniques, including Gray-Level Co-occurrence Matrix (GLCM), Histogram of Oriented Gradients (HOG), and Local Binary Patterns (LBP). The experimental results show that the SVM classifier trained with the proposed feature set achieves an accuracy of 82% and outperforms the conventional handcrafted feature extraction approaches. The dataset used includes only the Ash gourd plant variety and exhibits geographic and temporal biases, as it primarily covers data from specific regions and periods potentially limiting the model's effectiveness. The proposed design paves the way for intelligent decision-support systems for precision agriculture and sustainable crop management while providing a scalable, comprehensible approach to nutrient deficiency identification.

1. Introduction

Global food security still depends on agriculture with crop production and quality under threat due to climate change, unpredictable weather, and soil degradation, all of which are unprecedented in modern farming [1]. Identification and balancing of nutrient deficiencies in a systematic way and understanding and implementation of their management is essential for optimal crop health [2]. The deficiency of primary macronutrients especially nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), sulfur (S) and essential micro nutrients including iron (Fe), zinc (Zn), boron and manganese (Mn) are vital for plant biochemical and physiological processes including photosynthesis, enzyme activation and protein synthesis [3,4]. Therefore, efficient nutrient management is essential for crop productivity, quality, and the sustainability of horticultural production [5].

Ash gourd (*Benincasa hispida*), a member of the cucurbitaceous family, is widely cultivated in tropical and subtropical regions for its nutritional and medicinal value. The pharmaceutical and biomedical sectors [6] utilize Ash gourd due to its therapeutic properties. The nutrient content has a great impact on the growth and productivity of

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Ash gourd. The physiological effects of nutrient deficiencies in cucurbit crops have been demonstrated in several studies [7]. Iron (Fe) deficiency affects plant physiology and nitrogen metabolism, suggesting a link between nutrient uptake and utilization [8]. Similarly, deficiency of magnesium (Mg) reduces carbohydrate translocation, resulting in reduced yield and impaired seed germination in Ash gourd [9]. These deficiencies are evident on the leaves of the plant and include chlorosis, necrosis, leaf deformation and stunted growth. These symptoms also tend to be subtle in the early stages and are difficult to identify manually, leading to delays. Overall food quality and agricultural productivity are affected by prolonged nutrient deficiencies and can severely impair plant growth, reduce crop yield and, in extreme cases, lead to plant mortality. The problem becomes more complex in large agricultural fields where manual monitoring is labour-intensive and delays diagnosis and corrective nutrient management. Thus, early detection of nutrient deficiency is important for healthy plant growth, secondary disease prevention, and improved crop productivity [10].

Photosynthesis-based physiological measurements combined with statistical analysis have been successfully employed for early detection of phosphorus (P) deficiency stress in cucumber plants [11]. The findings stress the significance of timely and precise identification of nutrient deficiencies to ensure the sustainable cultivation of cucurbit crops. Recent studies on cucurbit nutrient deficiencies leverage datasets obtained from laboratory conditions, which limits the model's applicability in real-world situations.

Disease and nutrient deficiencies further impede human inspection. Agricultural experts identify nutrient deficiencies in ash gourd plants through manual inspection, soil testing and plant tissue analysis. These traditional techniques are labour-intensive, require field experience, and the presence of multiple leaves and background noise further amplify the challenges [12]. Rapid methods of testing soil and plant nutrients provide timely intervention and improved nutrient management strategies [13].

Computer vision techniques have been employed to detect and classify nutrient deficiencies in vegetable crops [14]. The detection of macro- and micronutrient deficiencies in cucurbit species has been greatly improved by recent developments in machine learning and deep learning, which have leveraged large datasets for better accuracy and consistency. Machine learning algorithms have been widely used to classify nutrient deficiencies [15]. Deep learning models have automated feature extraction, detection, segmentation, and classification, which are usually difficult and time-intensive [16]. Despite this, deep learning models often require large datasets [17], additional training time, and GPU-based computational resources, and they lack explainable feature analysis, making them more challenging to understand and necessitating larger convolutional networks.

On the other hand, machine learning models trained on handcrafted features are computationally efficient, require less training time and are better suited for deployment on edge devices and embedded systems [18]. Feature extraction methods, along with selective aggregation of features that leverage color, geometric shapes, and texture descriptors such as GLCM, HOG, and LBP, are widely utilized machine learning approaches. These features are classified using algorithms like Decision Trees, Random Forests, SVM, and Gradient Boosting. While these approaches are effective, they tend to learn generic visual features and often fail to capture domain-specific visual clues of nutrient deficiency, such as leaf vein patterns and physiologically relevant rule-based features. Incorporating these domain-specific characteristics can provide a more comprehensive representation of nutrient deficiency symptoms and improve classification performance.

In the proposed work, nutrient deficiencies are unveiled through patterns of color, vein, texture and spatial distribution of symptoms. Among these, vein patterns provide critical diagnostic information, as many deficiencies produce distinct interveinal chlorosis that varies according to the affected nutrient. These visual cues are complemented by rule-based methods that incorporate expert knowledge in the form of established physiological characteristics of nutrient deficiencies. Spatial features capture the positional distribution of these symptoms and thus distinguish localized interveinal chlorosis from diffuse or uniformly distributed yellowing. Texture descriptors measure the contrast variations caused by chlorotic mesophyll, while retaining the green vascular tissues. Color characteristics are used to discriminate between types of deficiencies, through identifying hue and saturation changes. Iron deficiency yields a bleached or light yellow with relatively low saturation. Magnesium deficiency has a distinct yellow hue and moderate saturation.

Therefore, integrating vein-based, rule-based, color, texture, and spatial features provide a more comprehensive representation of deficiency symptoms. The proposed AshGdNutDefAM framework provides an interpretable and discriminative representation of nutrient deficiency symptoms. These aggregated feature set is subsequently classified using machine learning algorithms to achieve accurate and reliable nutrient deficiency classification.

The key contributions of the proposed study are:

1. A feature extraction framework, includes fifty-five leaf characteristics, which integrates color analysis, texture assessment, vein pattern examination, rule-based method and spatial attribute evaluation of leaf images.
2. A hybrid methodology is employed, which successfully combines vein-based feature analysis to identify nutrient deficiencies, rule-based feature engineering by an expert botanical understanding, and assessment via data-driven machine learning classifiers.
3. Comparative analysis of conventional machine learning classifiers, like Random Forest, SVM, and Gradient Boosting, to evaluate the best classifier.
4. Compilation and curation of a dataset of magnesium and iron nutrient deficiencies in Ash gourd plants in-field environment, to address the lack of publicly accessible data for this particular crop and deficiency types.

2. Related works

Several studies have employed computer vision techniques, machine learning, deep learning-based algorithms, and feature aggregation for nutrient deficiency detection and some of the major contributions are summarized below.

Aleksandrov et al. [19] proposed a method based on rapid chlorophyll fluorescence for the detection of nutrient deficiencies in plants. An ANN was trained to identify deficiencies in bean plants for N, P, K, Ca, or Fe. Shi et al. [20] established a diagnostic approach for nitrogen, magnesium and potassium deficiencies in cucumber leaves based on chlorophyll density with a nutrient identification rate of 96.67%. Peng et al. [21] studied computer vision techniques, which have emerged as a transformative approach in crop nutrition monitoring by providing a non-destructive, effective way for plant nutrient evaluation. Barbedo et al. [22] investigated the detection of nutrition by analyzing various feature extraction techniques, including color features, texture descriptors, and morphological characteristics, in combination with classical machine learning classifiers, using proximal images and achieved 80–90% accuracy in crop type classification. Rahadiyan et al. [23] developed a feature aggregation technique combined with machine learning to identify nutrient deficiencies in chilli plants. This involved color features of HSV and Lab color spaces, texture features using Gray Level Co-occurrence Matrix (GLCM), and shape descriptors. CNN, SVM, and MLP achieved accuracies of 97.76%, 90.55%, and 89.70%, respectively.

Farzizadeh et al. [24] proposed a smart nutrient-detection system using machine learning and fusion techniques to classify leaf images into healthy, potassium-deficient, nitrogen-deficient, and phosphorus-deficient classes, achieving an accuracy of 98.17%. Pinheiro et al. [25] proposed an aggregation strategy of texture and multiscale

shape features that improved the classification accuracy of leaf images using the random forest classifier. Rathnayake et al. [26] worked on the identification and categorization of macronutrient deficiencies such as nitrogen, phosphorus, and potassium in rice plants using pixel ratio analysis in the HSV color space, which resulted in 90% accuracy. Patel et al. [27] investigated nutrient deficiencies, such as zinc, nitrogen, phosphorus, and potassium, using machine learning algorithms. The features were extracted using DWT, GLCM, Gabor, and LBP, and were classified using SVM, KNN, DT, and GB. SVM with RBF showed a precision of 71% and 67%, a recall of 65%, and an F1 score of 65%, with an accuracy of 67%. Shadrach et al. [28] developed an optimal nutrient-deficiency classification model using transfer learning for ridge gourd plant variety. The study proposed a unique ring-toss game to classify nitrogen, phosphorus, potassium, calcium, and iron deficiencies. Pre-processing, feature extraction, segmentation, hyperparameter tuning, and classification were performed using the RTGODTL-NDC technique.

Nabil et al. [29] proposed Nutrispace, a color space designed to enhance deep learning-based early detection of nitrogen and potassium deficiencies in bitter gourd, Ash gourd, and snake gourd. Sakthipriya et al. [30] proposed an approach that combines Convolutional Neural Networks with a Genetic Algorithm (GA) for nutrient management in rice crop. Nitrogen, phosphorus, and potassium were considered in the proposed genetic algorithm (GENCNN), and an accuracy of 99% was observed. Sirsat et al. [31] studied wheat leaf images to identify nutrient deficiencies. They used sixteen feature vectors, comprising statistical, grey-level texture features based on the co-occurrence matrix, color features, local binary pattern features, and feature vectors combining color and grey-level texture features. Classification was achieved using a radial basis function (RBF) neural network with color features (vector fosRGB), yielding a kappa score of 57% and an accuracy of 81.2%.

Nadafzadeh et al. [4] developed an approach to identify nutrient deficiency stress in spinach for iron, zinc, and manganese using computer vision. The study employed a combination of color indices, RGB, HSV, and Lab and twenty-two GLCM texture features. The classification was performed using an ANN, an SVM, and a Random Forest. ANN achieved the highest accuracies of 84% for Mn, 88% for Fe, and 76% for Zn, respectively. Rahadiyan et al. [32] designed an intelligent hydroponics system based on color, shape, and texture to identify macronutrient deficiency in intelligent chili hydroponics systems. Bera, A. et al. [33] proposed a graph convolutional network-based classification approach, PND-Net, for identifying nutrient deficiency in banana, coffee, and potato plants using a public dataset. Sathyavani, R et al. [34] proposed an image acquisition-based classification model that utilizes Internet of Things (IoT) for data acquisition and convolutional neural networks (CNN), DenseNet-BC, for the task of classification, and results show that the proposed method obtains improved classification accuracy in terms of accuracy and F-measure than other deep learning models.

From the literature survey, the following research gaps are noted

- Nutritional deficiency identification in vegetable crops has mainly relied on soil and plant tissue analysis, and it is a time-consuming process and less adaptable to real-life scenarios throughout crop management.
- Traditional methods rely on generic features including GLCM, LBP, HOG, color histograms and lack domain knowledge, which is critical in nutrient deficiency identification.
- Most studies utilize small, controlled, or synthetic datasets with limited real-world variability in terms of lighting conditions, growth stages, and deficiency severity levels. No publicly available, crop-specific benchmark dataset exists for nutrient-deficiency detection, making cross-study comparisons difficult. Studies have focused on datasets with plain backgrounds and lower complexity, which pose challenges in field settings.
- There is limited research work on nutrient deficiency identification and detection in Ash gourd. Existing models have limited generalization across different crop species. Nutritional deficiencies in Ash gourd plants are closely associated with image color, vein, texture, and domain-specific knowledge, which existing methods lack.
- Most of the studies focused on the detection of primary nutrient deficiencies, particularly nitrogen(N), phosphorus(P), and potassium (K), while secondary and micronutrient deficiencies such as iron, magnesium, manganese, zinc, and boron are underexplored.
- The detection of multi-nutrient deficiency for plant crops poses challenges in-field scenarios, and this necessitates the development of machine learning models suitable for in-field nutrient deficiency diagnosis.

3. Methodology

AshGdNutDefAM, a hybrid feature-based machine learning system for the automatic detection of nutrient deficiencies in Ash gourd (*Benincasa hispida*) plants, has been proposed as part of the study. Observations revealed that traditional diagnostic techniques such as manual visual inspection and laboratory chemical analysis are inherently labour-intensive and expensive, and prone to human error and subjectivity. The AshGdNutDefAM framework, as depicted in Fig. 1, integrates specialized botanical knowledge of vein patterns and rule-based feature engineering with the classification capabilities of traditional machine learning classifiers. The dataset utilized in this study consists of Ash gourd plant images exhibiting nutrient deficiencies, encompassing magnesium (Mg) deficiency, iron (Fe) deficiency, and healthy.

This framework involves image acquisition, pre-processing, background removal, multi-domain feature extraction, and machine-learning classification to categorize images into three distinct classes: magnesium (Mg) deficiency, iron (Fe) deficiency, and healthy. During these phases, image preprocessing is performed, and unique features such as color, texture, shape, vein, and rule-based features are extracted to represent nutrient deficiency indicators effectively. The identified characteristics are grouped using Random Forest, SVM and Gradient Boosting. This offers a scalable, cost-effective, and straightforward approach to rapidly recognize nutrient deficiencies, thus facilitating precision fertilization approaches for improving crop productivity and overall plant health.

3.1. Image acquisition and preparation

The dataset of 500 images of the Ash gourd plant variety was collected from a two-acre plantation in Kiralu, Mysore District, Karnataka, India (Latitude: 12°13′56.57″ North, Longitude: 76°44′39.04″ East) during the monsoon season in July. Images were taken 30 days after the crops were planted in red soil on prepared dry land. Each image, with a resolution of 3060 × 4080 pixels and featuring one to five leaves, was captured with an iQOO smartphone running Android and equipped with a 50 MP camera.

The images were captured includes both background and foreground leaves; some leaves are partially occluded or overlapping. The dataset provides an extensive representation of natural agricultural environments, with complex background elements, including soil, polythene covers, human interference, shadows, weeds, and scattered leaves throughout the frame. Images were captured at a distance of 1–1.5 feet, and the camera was zoomed to record deficiency symptoms on leaves in the foreground, background, and overlapping regions, enabling accurate annotation of the deficiency regions. The image samples were manually pre-processed to correct imperfections. Figs. 2 and 3 present sample images of magnesium, iron deficiency and healthy plant.

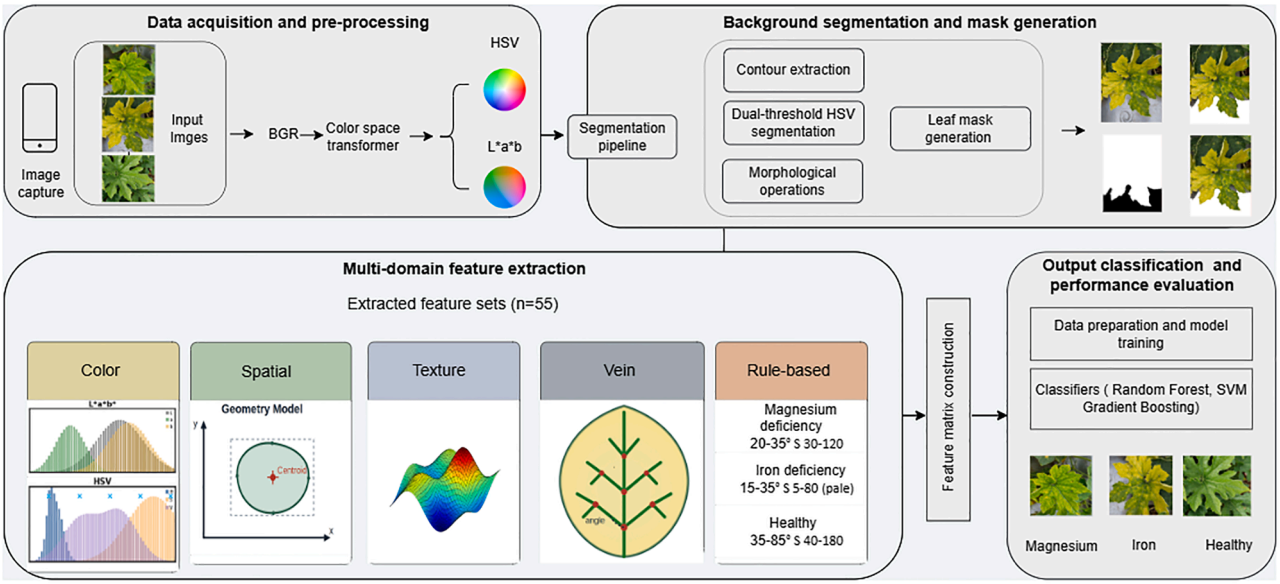


Fig. 1. Overall architecture of the proposed AshGdNutDefAM framework for nutrient deficiencies classification, illustrating image acquisition, preprocessing, feature extraction, and machine learning-based classifiers.

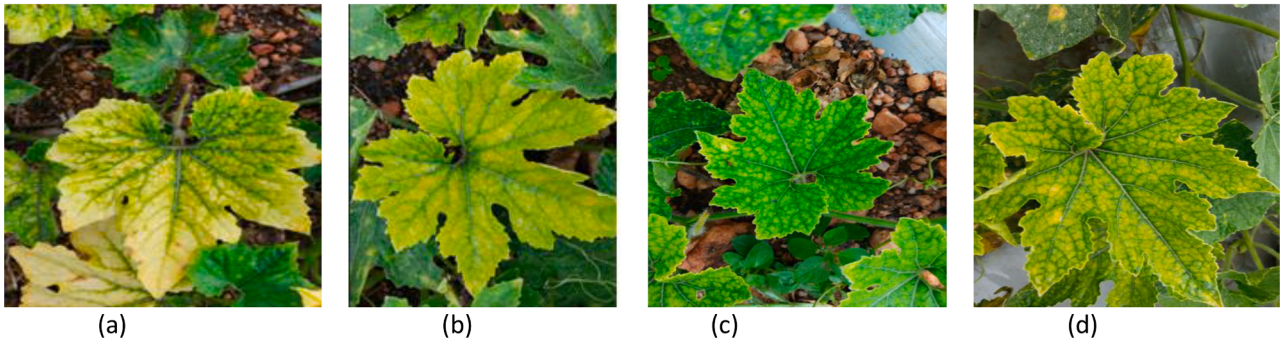


Fig. 2. Sample images of an Ash gourd plant variety captured under in-field conditions (a) (b) Iron deficient and (c) (d) Magnesium deficient.

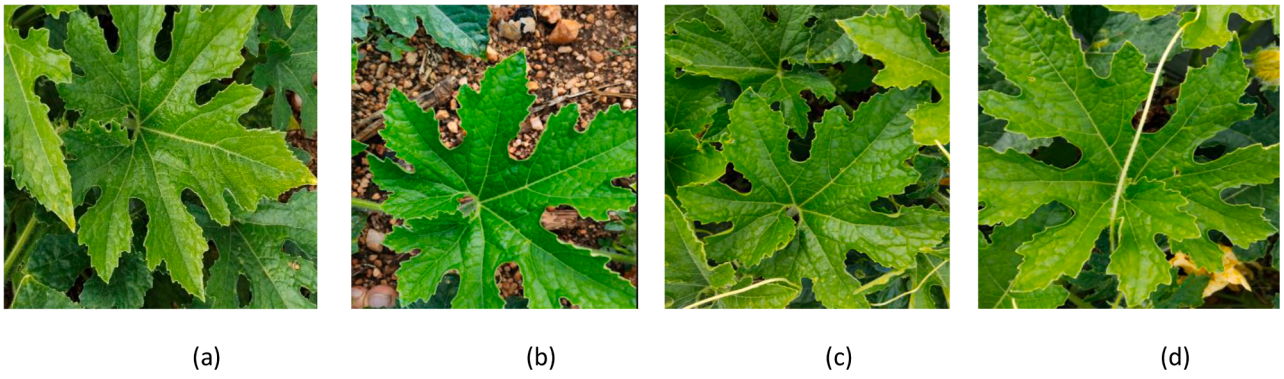


Fig. 3. (a) (b) (c) (d) Sample images of healthy Ash gourd plant variety captured under in-field conditions.

3.2. Image pre-processing

3.2.1. Color space transformation and dual-threshold segmentation

The input images are subjected to leaf region segmentation to ensure that the feature extraction process is applied to leaf tissue by excluding background artefacts. Eq (1) denotes an input image with three channels, Blue, Green, and Red, for dimensions of $h \times w \times 3$, where h and w denote the height and width, respectively.

$$I(x,y) = \mathbb{R}^{h \times w \times 3} \tag{1}$$

where (x, y) are pixel coordinates, $x \in \{1, 2, ..W\}$ and $y \in \{1, 2, ..H\}$ each pixel $I(x,y) = [B, G, R]$ represents the blue, green and red intensity values.

The HSV (Hue, Saturation, Value) and L*a*b color spaces are derived from the BGR image to provide better color discrimination of plant tis-

sue compared to BGR, as shown in Eq (2).

$$I_{HSV(x,y)} = F_{BGR\to HSV}(I(x,y))\ I_{Lab(x,y)} = F_{BGR\to LAB}(I(x,y)) \quad (2)$$

Two complementary binary masks, $M1$ *and* $M2$, for healthy green and yellow regions, which are indicative of chlorosis symptoms, are generated using adaptive thresholding as depicted in eq's (3) and (4).

$$M1(x,\ y) = 1(25 \le H \le 90) \wedge (20 \le S \le 255) \wedge (20 \le V \le 255) \quad (3)$$

$$M2(x,\ y) = 1(15 \le H \le 35) \wedge (20 \le S \le 255) \wedge (20 \le V \le 255) \quad (4)$$

where $H \in [0,\ 179]$, $S,\ V \in [0,\ 255]$, and $1(\cdot)$ is the indicator function that returns 1 when the condition is true and 0 otherwise. A leaf region mask is created by combining $M1$ and $M2$ using a bitwise OR operation, as depicted in eq (5).

$$M_{combined} = M1_{x,y}\ V\ M2_{x,y} \quad (5)$$

3.2.2. Morphological refinement and background removal based on contour

Morphological refinement is carried out on the combined binary mask *Mcombined* to remove noise artefacts, fill small gaps within the leaf region, and produce a clean, continuous leaf boundary as depicted in Eq (6). Opening operations, represented by $\ominus$, eliminate isolated noise pixels and smooth the leaf contour. Closing operations are represented by $\oplus$, which connects nearby leaf regions and fills internal holes. A 5 × 5 structuring element K is applied in sequence.

$$Mrefined = (Mcombined \oplus K) \ominus K \quad (6)$$

After morphological refinement, contour extraction is performed that identifies all connected components in the mask. The final leaf is constructed by extracting the largest connected contour, which corresponds to the primary leaf region as depicted in Eq (7). Contour-based selection eliminates background artefacts, shadow regions, and any spurious detections.

$$M_{final} = argmax_{C\in\mathscr{C}}\ Area\ (C) \quad (7)$$

where C represents the set of all detected contours. The subsequent feature extraction operates on the segmented leaf tissue, with complete background removal. Let $\mathscr{L}$ represents the set of all leaf pixels after background removal, as given in Eq (8).

$$\mathscr{L} = \{(x,y) \mid M_{final}\ (x,y) = 1\} \quad (8)$$

3.3. Multi-domain feature extraction

The leaf image analysis commonly includes structural, texture, and color properties as common key features. In the proposed study rule-based features and vein features are aggregated with color, texture, and spatial features resulting in a unified feature descriptor vector. A comprehensive set of fifty-five features is extracted from the isolated leaf region by removing the background. Further, the analysis was conducted to explore the relationship between various feature extraction methods. Table 1 provides a comprehensive set of features extracted to represent low-level visual properties and high-level domain-specific characteristics of nutrient deficiencies.

Table 1
Feature representation of the proposed framework across five categories.

Category	Count	Features
Color	20	H_mean, H_std, H_median, H_q25, H_q75, S_mean, S_median, S_std, S_q25, S_q75, V_mean, V_median, V_std, V_q25, V_q75, L_mean, a_mean, b_mean, yellow_ratio, green_ratio
Spatial	15	leaf_area, perimeter, compactness, aspect_ratio, extent, top_bottom_H_diff, top_bottom_S_diff, top_bottom_V_diff, left_right_H_diff, left_right_S_diff, left_right_V_diff, quadrant_H_variance, quadrant_S_variance, quadrant_V_variance, spatial_uniformity
Texture	9	gray_mean, gray_std, gray_entropy, edge_density, texture_uniformity, gradient_mean, gradient_std, laplacian_variance, laplacian_mean
Vein Pattern	8	vein_density, vein_nonvein_H_diff, vein_nonvein_S_diff, vein_nonvein_V_diff, interveinal_chlorosis_score, vein_segments, avg_vein_length, vein_continuity
Rule-based	3	Mg_rule_score, Fe_rule_score, Healthy_rule_score
The pseudocode of the proposed framework is summarized below		
		Input: RGB image I
		Output: Extracted color, spatial, texture, vein and rule-based features
1		Convert input image from BGR to HSV and LAB Color space conversion
		$I_{hsv} \leftarrow cvtColor(I, BGR)$
		$I_{Lab} \leftarrow cvtColor(I, BGR)$
2		Perform segmentation to remove background noise
		$M \leftarrow ExtractionLeafMask(I)$
3		*If* $\lvert M \rvert < 100$ *then return null;*
		Mask too small, invalid leaf region
4		*Else*
5		$P_{hsv} \leftarrow pixels\ of\ I_{hsv}\ where\ M > 0$
6		$P_{Lab} \leftarrow pixels\ of\ I_{Lab}\ where\ M > 0$
7		$P_{BGR} \leftarrow pixels\ of\ I\ where\ M > 0$
8		$f_{Color} \leftarrow ExtractColorFeatures\ (P_{hsv},\ P_{Lab},\ P_{BGR})$
9		$f_{Spatial} \leftarrow ExtractSpatialFeatures\ (M,\ I_{hsv})$
10		$f_{Texture} \leftarrow ExtractTextureFeatures\ (I,\ M)$
11		$f_{Vein} \leftarrow ExtractVeinFeatures\ (I,\ M, I_{hsv})$
12		$f_{Rule-based} \leftarrow ExtractRuleScores\ (I_{hsv},\ M)$
13		$return\ f \leftarrow \left[f_{color} \,\middle\|\, f_{Spatial} \,\middle\|\, f_{Texture} \,\middle\|\, f_{Vein} \,\middle\|\, f_{Rule-based}\right]$

3.3.1. Color feature extraction

Color is an important trait in a leaf that gives valuable information about nutrient deficiencies. Each color channel shows specific information, so the channels were examined separately. This analysis highlights the unique characteristics of each channel. Twenty color features are extracted across different color spaces to characterize the chromatic properties of leaf tissue. This color space approach provides both perceptually uniform representations CIE $L*a*b*$ and Hue-based discrimination (HSV). The feature set includes fifteen statistical descriptors derived from the HSV color space: mean (μ), median, standard deviation (σ), first quartile $(Q25)$, and the third quartile $(Q75)$ as depicted in Eq (9) for each of the three channels $(H,\ S,\ V)$ which capture the central tendency, variability, and distribution shape of color values within the leaf. In addition, three mean values are extracted from the CIE $L*a*b*$ color space, consisting of the lightness channel $(L*)$, the green–red chrominance axis $(a*)$, and the blue–yellow chrominance axis $(b*)$.

$$\mu_c = \frac{1}{N}\sum_{(x,y)\in\mathscr{L}} C(x,y), \quad \sigma_c = \sqrt{\frac{1}{N}\sum_{x,y\in\mathscr{L}} C(x,y) - \mu_c)^2} \quad (9)$$

Two color distribution ratios are calculated to quantify the relative proportions of green and yellow pixels as depicted in Eq (10), providing discriminative measures for assessing leaf health and identifying chlorosis patterns.

$$R_{yellow} = \frac{1}{N}\sum_{(x,y)\in\mathscr{L}} 1\ (18 \le \mathrm{H}\ (\mathrm{x,y}) \le 35),\ R_{green} = \frac{1}{N}\sum_{(x,y)\in\mathscr{L}} 1\ (35 \le \mathrm{H}\ (\mathrm{x,y}) \le 85) \quad (10)$$

3.3.2. Spatial feature extraction

Fifteen spatial features are extracted, including core morphological descriptors such as leaf geometric morphology and spatial heterogeneity of color patterns, which define the leaf's physical structure, and spatial color distribution metrics that identify localized deficiency patterns as depicted in Eq (11). Spatial features enable quantification of color variance across the leaf surface and capture patterns of interveinal

chlorosis, tip burn, and marginal necrosis, which are critical features of nutrient deficiencies.

$$A_{leaf} = \frac{4\pi A_{leaf}}{P^2 + \varepsilon} \tag{11}$$

where P is the contour perimeter, C is the compactness measure ranging from 0 to 1, with 1 indicating a perfect circle, and $\varepsilon = 10^{-6}$ prevents division by zero. Additional shape descriptors include aspect ratio, which indicates the width-to-height ratio of the bounding box, and extent, the ratio of leaf area to bounding box area. To assess spatial color heterogeneity and detect non-uniform nutrient deficiency symptoms, the leaf region is divided into top-bottom and left-right quadrants, and color statistics are computed for each partition, as shown in Eq (12). Metrics are also computed for saturation (S) and value (V) channels across different spatial regions.

$$\Delta \mathrm{H_{tb}} = \left| \mu_{H_{top}} - \mu H_{bottom} \right|, \ \Delta \mathrm{H_{lr}} = \left| \Delta_{H_{left}} - \mu_{H_{right}} \right| \tag{12}$$

3.3.3. Texture analysis with gradient-based and statistical descriptors

Texture features are visual patterns that show changes in pixel brightness, giving details about the surface qualities of an image. A set of nine texture features from leaf surface morphology and tissue structure is extracted. Grayscale conversion is performed using the standard luminance-weighted transformation, as depicted in Eq (13).

$$I_{gray} = 0.299R + 0.587G + 0.114B \tag{13}$$

First-order texture statistics, mean, and standard deviation are computed from the grayscale intensity distribution within the leaf region. Shannon entropy indicates information-theoretic complexity, with $p(i)$ corresponding to the normalized histogram probability for the intensity level i and is depicted in Eq (14).

$$H_{gray} = -\sum_{i=0}^{255} p(i) Log_2(p(i) + \in) \tag{14}$$

Gradient-based texture descriptors are obtained through the usage of Sobel operators in the horizontal (Gx) and vertical (Gy) orientations. Gradient magnitude reflects local intensity variations, as illustrated in Eq (15). The Laplacian operator ($\Delta^2 I$) checks smooth texture which is sensitive to quick changes in intensity. The properties of these textures are characteristic of a smooth, uniform texture of healthy leaves which are distinct from the irregular, distorted patterns of tissue missing nutrients.

$$G_{Mag} = \sqrt{\left(G_x^2 + G_y^2\right)}, \quad \Delta^2 I = \frac{\partial^2 I}{\partial^2 x^2} + \frac{\partial^2 I}{\partial^2 y^2} \tag{15}$$

3.3.4. Vein features

Eight vein-pattern characteristics are used to detect interveinal chlorosis caused by insufficient magnesium and iron supply. In the deficiency conditions, chlorophyll breaks down in the interveinal mesophyll tissue, and the veins of the vascular tissue remain green, creating a characteristic reticulated pattern. Canny edge detection and morphological dilation are applied together to delineate vein regions as depicted in Eq (16).

$$M_{vein} = Dilate\left(Canny\left(I_{gray}, 30, 100\right), K_{3\times 3}\right) \wedge M_{final} \tag{16}$$

where Canny (·, 30, 100) denotes edge detection with lower threshold 30 and upper threshold 100, and $K_{3\times 3}$ is, a 3 × 3 morphological structuring element used for dilation to make veins more closely connected. The vein density, which refers to the proportion of vein pixels compared to the total leaf area, is depicted in Eq (17).

$$D_{vein} = \frac{\sum_{x,y} M_{vein}\ (x,y)}{A_{leaf} + \in} \tag{17}$$

The colorimetric difference between vein and non-vein areas is the most vital indication of interveinal chlorosis. The interveinal chlorosis score is created to determine the unique pattern in which veins stay green with high hue values, while interveinal tissue yellows with low hue values and less saturation, and Eq. (18) depicts this

$$S_{chlorosis} = \left| \mu_{Hvein} - \mu_{H_{non-vein}} \right| \times \left(1 - \frac{|\mu_{Svein} - \mu S_{non-vein}|}{255}\right) \tag{18}$$

3.3.5. Rule-based probability scores

Rule-based probability scores consist of three scores that encode expert knowledge of characteristic nutrient-deficiency symptoms. These scores transform the visual diagnostic criteria into quantitative features suitable for computation. The magnesium deficiency score indicates that pixels match the interveinal chlorosis pattern, with yellow color and moderate saturation, as shown in Eq. (19). The rule uses the yellow-green color (hue 20–35°), medium saturation (30–120), and a brightness score $V \geq 80$ that are common in interveinal tissue that lacks the necessary magnesium.

$$S_{Mg} = \left(\frac{1}{A_{leaf}}\right) \sum_{(x,y)\in \mathscr{L}} 1\ ((20 \leq H \leq 35) \wedge (30 \leq S \leq 120) \wedge (V \geq 80)) \tag{19}$$

The iron deficiency score reflects the pale yellow-to-white discoloration and low saturation values characteristic of severe iron chlorosis, specifically in young leaves, as shown in Eq. (20). The rule identifies pale yellow regions (hue 15–35°) with low saturation (5–80) and high brightness ($V \geq 150$), suggesting severe iron deficiency.

$$S_{Fe} = \frac{1}{A_{leaf}} \sum_{(x,y)\in \mathscr{L}} 1\ ((15 \leq H \leq 35) \wedge (5 \leq S \leq 80) \wedge (V \geq 150)) \tag{20}$$

The healthy score measures the percentage of pixels with a uniform green color and moderate brightness, as shown in Eq. (21). Balanced brightness (60–200) with moderate to high saturation (40–180) for a range of green hues (35–85°) is covered by this rule. Data-driven statistical attributes are enhanced by three rule-based metrics by offering transparent, comprehensible features.

$$S_{healthy} = \left(\frac{1}{A_{leaf}}\right) \sum_{(x,y)\in \mathscr{L}} 1\ ((35 \leq H \leq 85) \wedge (40 \leq V \leq 200) \wedge (60 \leq V \leq 200)) \tag{21}$$

3.3.6. Feature matrix and normalization

A structured feature matrix for machine-learning classifiers is created from features extracted from the leaf images. Twenty color features, fifteen spatial features, nine texture features, eight vein characteristics, and three rule-based features comprise the fifty-five-dimensional feature vector of a single leaf image, represented by each row. The corresponding label vector encodes the nutrient status class as depicted in Eq. (22).

$$X \in \mathbb{R}^{n\times 55}, \ y \in \{0, 1, 2\}^n \tag{22}$$

where n is the total number of leaf images in the dataset, X is the feature matrix, and y encodes the class labels as 0 for Healthy, 1 for magnesium deficiency, and 2 for iron deficiency. To ensure that all features contribute equally to classification and to prevent features with larger numerical ranges from dominating the learning process, z-score normalization is applied to the feature matrix as depicted in Eq. (23).

$$X_{Scaled} = \frac{X - \mu X}{\sigma_X} \tag{23}$$

where μX and σ_X represent feature-wise mean and standard deviation vectors, which are computed exclusively on the training data to prevent data leakage. Using stratified random sampling, the dataset is split into 80% training and 20% testing subsets. This approach maintains class balance between the two sets, which is essential for a balanced assessment.

3.4. Machine learning classification

Classification refers to a machine learning technique that is used to classify input data into certain classes or labels by learning the patterns from the training data. The model provides automated decision-making by predicting the classes of new unseen data points. In this study, the classification task was carried out using three different methods: Random Forest, Support Vector Machine with an RBF kernel, and Gradient Boosting, which were chosen based on their complementary strengths in handling high-dimensional and heterogeneous feature sets that belongs to color, texture, vein, rule-based and spatial features. Random Forest provides ensemble-based robustness against individual feature noise; SVM with RBF kernel offers strong generalization in non-linearly separable feature spaces; and Gradient Boosting provides a sequential error-correction mechanism that is particularly effective when feature contributions are unequal across classes that are discussed below:

3.4.1. Random forest ensemble classifier

Random forest selects the class with the most predictions across all trees and trains on multiple decision trees. At each node split, only a random subset of features is considered when training the tree on a sample of training data. The randomization approach maintains high predictive accuracy while reducing overfitting. The random forest classifiers consider 200 decision trees with a maximum depth of 15, a minimum samples-per-split of 5, and a minimum samples-per-leaf of 2. The prediction for a sample x is given by Eq. (24)

$$\hat{y} = mode\ \{T_1(x),\ T_2(x), \ldots T_{200}(x)\} \tag{24}$$

3.4.2. SVM with RBF kernel

SVM is a maximum-margin classifier that locates the best hyperplane in a transformed feature space to separate different classes. For non-linearly separable data, the kernel trick maps the input features into a higher-dimensional space where linear separation becomes possible. The SVM considers the Radial Basis Function (RBF) kernel, which can model complex, non-linear decision boundaries and the optimization problem for multi-class classification is depicted in Eq. (25).

$$\begin{aligned} &Minimize\ \left(\frac{1}{2}\right)\ |w^2| + C\sum_i \xi_i \\ &subject\ to\ y_i(w{\cdot}\varphi(x_i + b)) \geq 1 - \xi,\ \xi_i \geq 0 \end{aligned} \tag{25}$$

3.4.3. Gradient boosting sequential ensemble

Gradient Boosting is an ensemble technique that builds a series of weak learners sequentially, with each new tree correcting the errors of the preceding ensemble. Gradient Boosting uses an additive model that iteratively minimizes a loss function via gradient descent in the function space and leverages a hundred boosting stages, a learning rate of 0.1, and a maximum tree depth of 5. The prediction for the sample x after M boosting iterations is depicted in Eq. (26).

$$F_{M(x)} = F_{M-1}(x) + \eta{\cdot}h_M(x) \tag{26}$$

3.4.4. Validation and training strategy

The training labels and standardized feature matrix generated are used to train three classifiers. Five-fold stratified cross-validation is carried out to assess the model's generalisation and identify overfitting on new data from the training set. The training data is partitioned into a sub-training set and a validation set within each fold, while maintaining class proportions across folds. The cross-validation accuracy is given by Eq. (27).

$$Accuracy_{cv} = \frac{1}{K}\sum_{k=1}^{K} Accuracy_k \tag{27}$$

where $K = 5$ is the number of folds and $Accuracy_k$ is, the classification accuracy on the kth fold. The cross-validation standard deviation quantifies model stability across different data partitions.

3.4.5. Evaluation metrics

The proposed framework is evaluated using Accuracy, Precision, Recall, and F1 Score metrics. TP stands for True Positives, where the model has accurately identified the instance of the class. TN stands for True Negatives, correctly identified instances of other classes. FP stands for False Positives, incorrectly predicted as the class but belonging to another class. FN stands for False Negatives, incorrectly predicted as another class when actually belonging to this class.

Precision is the proportion of correct predictions among all instances predicted to belong to a particular class, measuring the classifier's accuracy. It is represented in Eq. (28).

$$Precision\ = \frac{TP}{(TP + FP)} \tag{28}$$

Recall is the proportion of actual class instances that are correctly identified, quantifying the classifier's completeness Eq. (29).

$$Recall\ = \frac{TP}{TP + FN} \tag{29}$$

The F1 Score is the harmonic mean of Precision and Recall. It is represented in Eq. (30).

$$F1\ Score = 2 * \frac{Precision\ \times Recall}{Precision\ \pm Recall} \tag{30}$$

Overall classification accuracy measures the proportion of correctly classified instances across all classes and is given by Eq. (31).

$$Accuracy\ = \frac{(TP\ +\ TN)}{(TP\ +\ TN\ +\ FP\ +\ FN)} \tag{31}$$

4. Experiments

4.1. Experimental environment

The AshGdNutDefAM framework was implemented in Python using libraries, namely OpenCV for image processing and computer vision operations, scikit-learn for preprocessing, evaluation, and machine learning, NumPy for numerical array operations and mathematical computations, Pandas for structured data manipulation and feature organization, and Matplotlib and Seaborn for comprehensive visualization of results. Experiments were conducted in Google Colab, consisting of Python 3.10, 12.7 GB of RAM, and Intel Xeon CPUs.

4.2. Results

In this section, experimental results are evaluated with three machine learning methods, namely Random Forest, SVM, and Gradient Boosting, for the three classes: Magnesium, Iron, and Healthy. Table 2 shows that SVM achieved the best overall accuracy of 0.82, compared to Random Forest with 0.77 accuracy and Gradient Boosting with 0.69 accuracy. The Iron class presented the best classification across Random Forest and SVM models, low performance with the Gradient Boosting

Table 2
Performance comparison of machine learning classifiers for classification of Healthy, Iron and Magnesium deficiency Ash gourd plant variety using proposed feature set consisting of Color, Texture, Shape, Vein and Rule-based features.

Feature set	Classifier	Class	Accuracy	Precision	Recall	F1-Score
Color + Spatial+ Texture+ Vein + Rule-based	Random Forest	Healthy	0.77	0.81	0.75	0.78
		Magnesium	0.76	0.71	0.75	0.73
		Iron	0.75	0.82	0.82	0.81
	Support Vector Machine	Healthy	**0.82**	**0.76**	**0.74**	**0.75**
		Magnesium	**0.81**	**0.78**	**0.76**	**0.77**
		Iron	**0.82**	**0.83**	**0.88**	**0.85**
	Gradient Boosting	Healthy	0.69	0.70	0.58	0.63
		Magnesium	0.68	0.63	0.70	0.66
		Iron	0.67	0.76	0.75	0.76

classifier. For the Healthy class, Random Forest yielded the best F1-Score of 0.78 compared to SVM and Gradient Boosting classifiers, whereas the Magnesium class showed F1-Scores of 0.73 for Random Forest, 0.77 for SVM, and 0.66 for the Gradient Boosting classifier. Moreover, the Iron class was the most consistently identified by all the models. The proposed hybrid feature-based machine learning framework demonstrated the most consistent and superior performance across three nutrient deficiency categories, effectively distinguishing between Magnesium, Iron, and Healthy categories.

4.2.1. *Feature visualization*

Fig. 4 represents feature visualization for the proposed method, which distinguishes the characteristics of each class across the four feature categories of healthy, magnesium-deficient, and iron-deficient classes. The spatial features describe the asymmetry of nutrient stress reflected in the leaf surface, with nutrient-deficient samples showing a more significant brightness differential across the leaf than healthy samples. The texture features highlight the surface loss homogeneity associated with nutrient stress, where healthy samples show lower entropy and higher uniformity, and nutrient-deficient samples present increased textural irregularity that reflects the chlorotic or necrotic patches on the leaf. The vein-based features indicate the propensity for iron-deficiency chlorosis to preserve vein pigmentation while the interveinal tissue yellows, creating a significant hue and saturation differential between vein and non-vein regions that is not present in healthy tissue. The rule-based scores translate these visual cues into interpretable, domain-informed indicators consistent with the qualitative symptoms used in field diagnosis, improving both the discriminative power and interpretability of the classifiers.

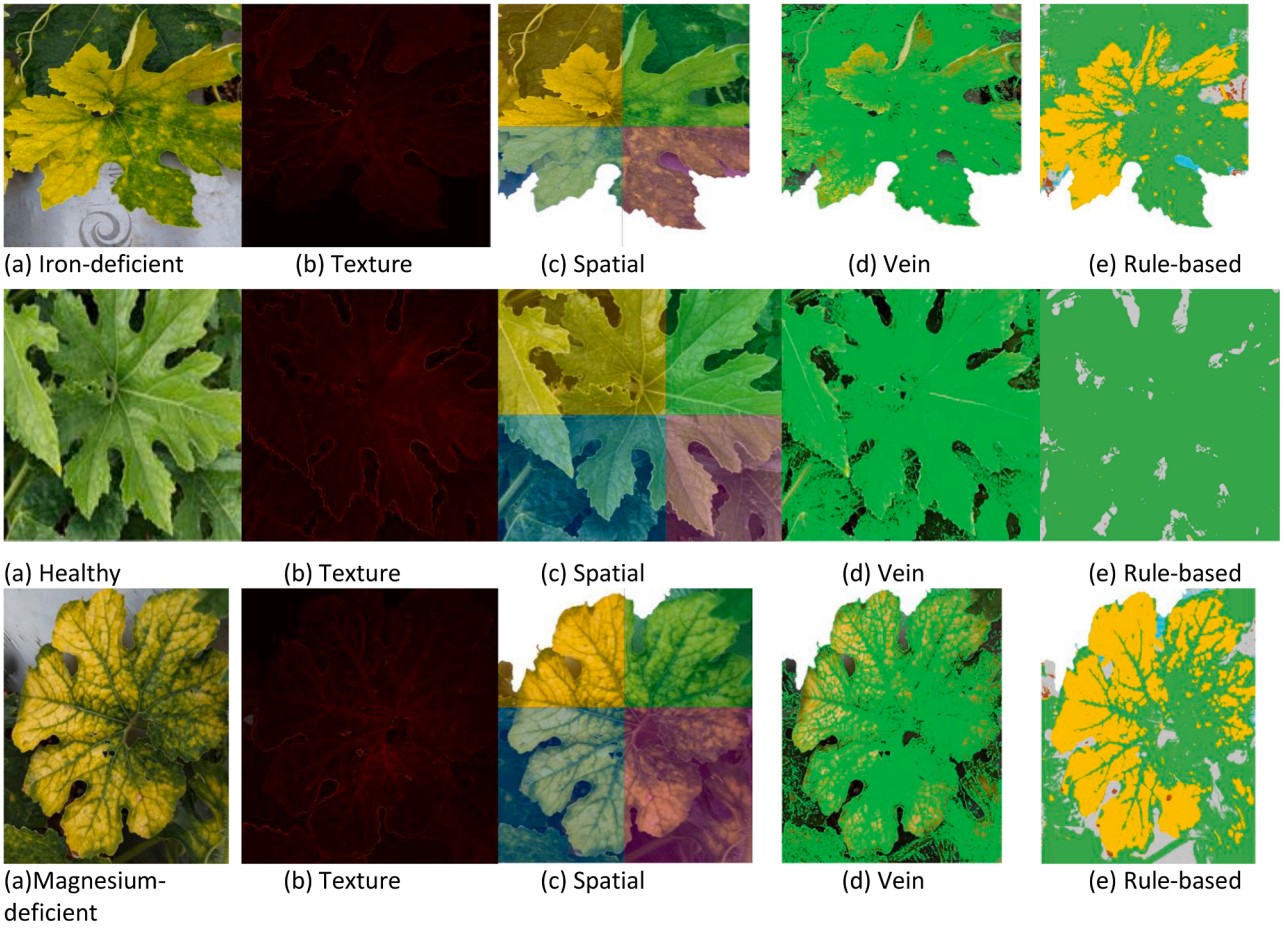


Fig. 4. Feature visulization of the proposed AshGdNutDefAM framework (a) Iron-deficient, Healthy, Magnesium-deficient original images (b) Texture,(c) Spatial,(d) Vein, (e) Rule-based representation.

4.2.2. Feature ranking

The analysis focused on exploring the relationships and dependencies between color, texture, spatial, vein, and rule-based features across the three machine learning classifiers. The feature correlations were examined to gain insight into the interaction and contribution of these features to the overall image characteristics, as shown in Fig. 5.

The observed divergence in feature reliance across classifiers suggests that different learning algorithms preferentially exploit different diagnostic cues, supporting the multi-model or ensemble approaches to capture discriminative information available across feature domains. The SVM model shows a clear dependence on vein-pattern features, especially vein_nonvein_H_diff and interveinal_chlorosis_score, suggesting it is more sensitive to local color variations between veins and the surrounding mesophyll. The rule-based probability scores in the feature representation reflect domain expert knowledge in the learning process, making the proposed framework more interpretable and diagnostically reliable. The gray_entropy, texture_uniformity, and S_q75 emerge as consistently important features, serving as indicators of progressive leaf tissue degradation associated with nutrient deficiency conditions.

The importance of yellow_ratio among tree-based models indicates that global leaf yellowing is an important early sign of nutrient stress, consistent with the traditional agronomic diagnostic criteria. The significance of spatial features also implies that nutrient deficiencies tend to cause spatially non-uniform discoloration patterns rather than persistent, diffuse yellowing, providing discriminative spatial cues in addition to color-based features. Texture features also quantify structural changes that occur as plant tissue degrades under nutrient stress and measure changes in leaf-surface homogeneity that are not captured by color statistics alone. These features collectively provide a robust and distinctive representation for classifying healthy, magnesium-deficient, and iron-deficient Ash gourd leaves under field conditions.

4.2.3. Ablation study

A systematic ablation analysis was performed to assess the contribution of each feature category by progressively integrating color, spatial, texture, vein and rule-based features and determine the optimal feature combination. This systematic evaluation quantifies the contribution of each feature category and examines the effectiveness of their integration in improving classification performance. The experimental results for one feature, two features, three features, four features, and the complete feature set are discussed in this section.

4.2.4. AshGdNutDefAM – single feature category

In the initial stage, each feature category was evaluated, and results are presented in Table 3. Of the five feature categories assessed, color features performed best, with both the Random Forest and SVM models achieving an accuracy of 0.73 and 0.72, while the Gradient Boosting model had a lower accuracy of 0.67. The rule-based features, consisting of three nutrient diagnostic scores Mg_rule_score, Fe_rule_score, and Healthy_rule_score, performed well as an individual category, achieving accuracies of 0.60 for Random Forest, 0.65 for SVM, and 0.59 for Gradient Boosting classifiers. These results show that the rule-based extraction methodology successfully identifies deficiency patterns, thereby achieving the most favorable performance-to-feature ratio across all single-category sets. Vein and texture features achieved accuracies of 0.55 to 0.61 across classifiers. This suggests that vein and texture features capture structural and chromatic relationships between vascular and mesophyll tissue that are not fully represented by color statistics alone. Spatial features performed very low, with accuracies of 0.50 for Random Forest, 0.53 for SVM, and 0.43 for Gradient Boosting. This means that the discriminative information of individual feature domains is not enough to classify healthy, magnesium and iron deficiency in Ash gourd leaves correctly across three classifiers.

4.2.5. AshGdNutDefAM – combination of two feature categories

Eight two-feature combinations were evaluated, across all possible pairings of the five feature domains, and Table 4 shows the results. The feature set consisting of color and spatial, which includes 35 features

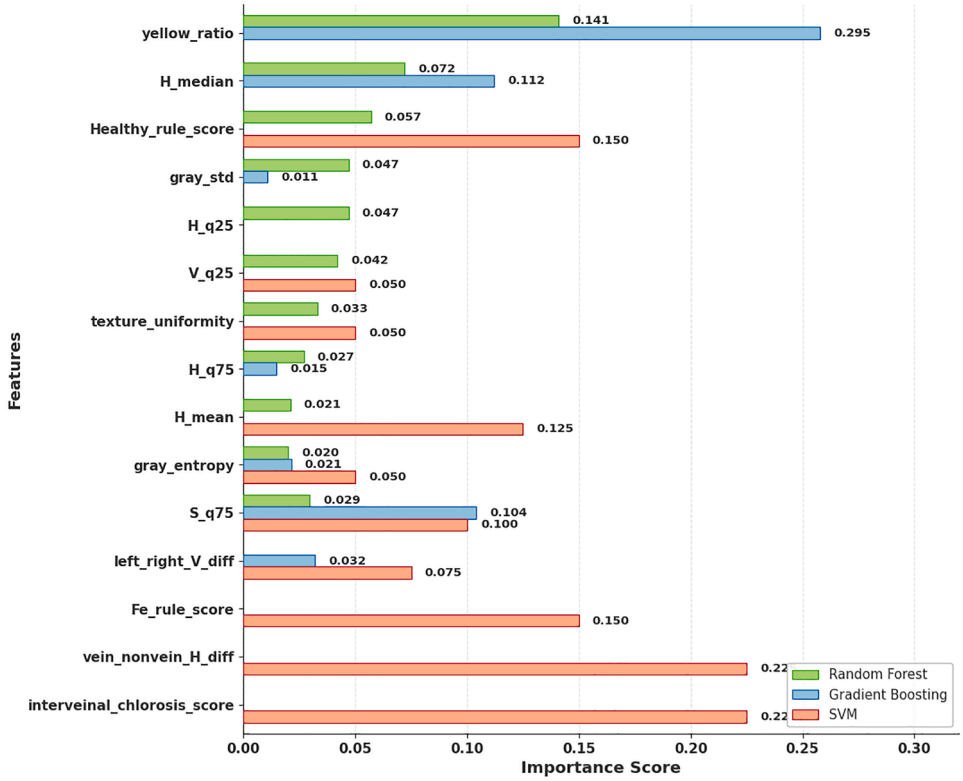


Fig. 5. A comparison of the top fifteen important features across the Random Forest, Gradient Boosting, and SVM classifiers.

Table 3
Performance of individual feature set of color, spatial, texture, vein, rule-based across random forest, SVM and gradient boosting classifiers.

Feature Set	Features	Random Forest				SVM				Gradient Boosting			
		Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
Color	20	0.73	0.74	0.72	0.71	0.72	0.75	0.71	0.73	0.67	0.65	0.63	0.64
Spatial	15	0.50	0.49	0.47	0.48	0.53	0.55	0.52	0.53	0.43	0.44	0.43	0.42
Texture	9	0.55	0.57	0.54	0.56	0.59	0.56	0.59	0.58	0.61	0.63	0.61	0.60
Vein	8	0.59	0.58	0.56	0.57	0.57	0.58	0.55	0.56	0.59	0.62	0.63	0.60
Rule	3	0.60	0.63	0.61	0.60	0.65	0.66	0.65	0.64	0.59	0.58	0.56	0.57

Table 4
Performance of possible two-feature combinations of color, spatial, texture, vein and rule-based features using random forest, SVM and gradient boosting classifiers.

Feature Set	Features	Random Forest				SVM				Gradient Boosting			
		Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
Color + Spatial	35	0.73	0.74	0.72	0.71	0.78	0.79	0.75	0.76	0.72	0.73	0.72	0.71
Color + Texture	29	0.71	0.73	0.71	0.72	0.73	0.75	0.72	0.73	0.71	0.73	0.71	0.72
Color + Vein	28	0.73	0.72	0.73	0.71	0.77	0.76	0.78	0.75	0.68	0.73	0.69	0.70
Color + Rule	23	0.76	0.75	0.74	0.74	0.72	0.75	0.74	0.75	0.67	0.67	0.68	0.66
Spatial + Texture	24	0.61	0.61	0.58	0.60	0.57	0.60	0.57	0.56	0.65	0.66	0.63	0.64
Spatial + Vein	23	0.49	0.48	0.49	0.47	0.53	0.54	0.51	0.52	0.51	0.51	0.48	0.50
Spatial + Rule	18	0.67	0.68	0.65	0.66	0.66	0.65	0.64	0.64	0.63	0.64	0.63	0.61
Texture + Vein	17	0.68	0.69	0.67	0.66	0.61	0.60	0.61	0.59	0.69	0.74	0.69	0.70
Texture + Rule	12	0.65	0.65	0.63	0.64	0.69	0.70	0.69	0.68	0.65	0.65	0.63	0.64
Vein + Rule	11	0.67	0.67	0.63	0.66	0.65	0.66	0.64	0.65	0.61	0.61	0.59	0.60

resulted in 0.73 accuracy for Random Forest, 0.78 for SVM, and 0.72 with Gradient Boosting. The feature set of color and vein, with 28 features, provided accuracies of 0.73 with Random Forest, 0.77 with SVM, and 0.75 for Gradient Boosting, showing vein-pattern features provide complementary diagnostic information when combined with color, reflecting the structural chromatic contrast between vascular and interveinal tissue that is characteristic of magnesium and iron deficiency. The color rule feature set of 23 features achieved the highest accuracy with results of 0.76 for Random Forest, 0.72 with SVM, and 0.67 for Gradient Boosting, suggesting that integrating domain-knowledge-driven rule scores with color statistics provides a particularly effective combination for the Random Forest classifier.

The lowest accuracies were achieved when combining spatial and vein features with 23 features, where Random Forest showed 0.49, SVM 0.53 and Gradient Boosting 0.52. Although performance was lower than with colour-inclusive combinations, the spatial and texture set of 24 features showed a moderate improvement, reaching 0.61 for Random Forest, 0.57 for SVM, and 0.65 for Gradient Boosting, suggesting that texture descriptors offer additional discriminative capacity over spatial features. The texture and vein with 17 features, attained accuracies between 0.61 and 0.66 across various classifiers. The texture and rule-based characteristics comprise 12 features, yielding an accuracy between 0.63 and 0.69, while the vein and rule-based set with 11 features attained an accuracy range of 0.61 to 0.67. The texture and rule-based set had the most robust performance, with the SVM attaining a score of 0.69. The accuracy of feature combinations without color features was lower, underscoring the significance of chromatic information as the primary discriminative characteristics for the classification of nutrient deficiencies in Ash gourd.

4.2.6. *AshGdNutDefAM – combination of three feature categories*

Three-category feature combinations from the five feature categories were tested, and results are shown in Table 5. The feature set with color, texture, and vein forming 37 features obtained the highest SVM accuracy of 0.78; Random Forest and Gradient Boosting achieved 0.71. A feature set consisting of color, spatial, and rule-based features with 38

Table 5
Performance of three-feature combinations of color, spatial, texture, vein, rule-based features using random forest, SVM and gradient boosting classifiers.

Feature Set	Features	Random Forest				SVM				Gradient Boosting			
		Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
Color + Spatial + Texture	44	0.73	0.73	0.70	0.72	0.77	0.78	0.76	0.75	0.69	0.69	0.67	0.68
Color + Spatial + Vein	43	0.76	0.74	0.76	0.75	0.75	0.74	0.76	0.73	0.73	0.74	0.71	0.72
Color + Spatial + Rule	38	0.75	0.75	0.77	0.74	0.76	0.75	0.73	0.74	0.78	0.77	0.76	0.76
Color + Texture + Vein	37	0.71	0.70	0.69	0.70	0.78	0.77	0.78	0.76	0.71	0.71	0.69	0.70
Color + Vein + Rule	31	0.74	0.76	0.75	0.75	0.76	0.75	0.76	0.74	0.72	0.71	0.68	0.70
Spatial + Texture + Vein	32	0.63	0.63	0.65	0.62	0.61	0.64	0.60	0.64	0.61	0.63	0.61	0.62
Spatial + Vein + Rule	26	0.73	0.76	0.73	0.74	0.67	0.68	0.68	0.67	0.71	0.73	0.71	0.72
Texture + Vein + Rule	20	0.71	0.72	0.71	0.70	0.71	0.73	0.71	0.72	0.67	0.68	0.67	0.66

features gained the best accuracy of 0.78; Gradient Boosting and Random Forest obtained 0.75. Color, spatial, and texture integration with 44 features achieved 0.77 for SVM, 0.73 for Random Forest, and 0.69 with Gradient Boosting. Color, spatial, and vein combination with 43 features resulted in accuracies of 0.76 for Random Forest, 0.75 for SVM, and 0.73 for Gradient Boosting. The color, vein, and rule feature set of 31 features recorded 0.74 with Random Forest, 0.76 with SVM, and 0.72 with Gradient Boosting.

The spatial, vein, and rule-based set with 26 features achieved the best results of 0.73 accuracy for Random Forest, 0.67 for SVM, and 0.71 for Gradient Boosting. The texture, vein, and rule-based combination with 20 features yielded 0.71 for Random Forest and SVM, and 0.67 for Gradient Boosting. However, the spatial, texture, and vein combination with 32 features had the lowest results for all three classifiers, with the accuracy ranging from 0.61 to 0.63. The three-feature category results show that performance continues to improve as color [35] is combined with progressively more feature domains.

4.2.7. AshGdNutDefAM – combination of four features analysis

To further investigate the contribution of each feature category, an ablation study was conducted, and the results are shown in Table 6. Spatial, texture, vein, and rule-based combination with 35 features resulted in accuracies of 0.71 for Random Forest, 0.69 for SVM and 0.66 for Gradient Boosting. Color, texture, vein, and rule-based combination with 40 features resulted in accuracies of 0.76 for Random Forest, 0.75 for SVM, and 0.67 for Gradient Boosting.

Color, spatial, vein, and rule-based integration with 46 features resulted in an SVM accuracy of 0.79, while Random Forest and Gradient Boosting obtained 0.78 and 0.71, respectively. Color, texture, spatial, and rule-based combination with 47 features resulted in accuracies of 0.77 for Random Forest, 0.78 for SVM, and 0.69 for Gradient Boosting. Color, texture, spatial, and vein combination with 52 features resulted in accuracies of 0.75 for Random Forest, 0.77 for SVM, and 0.76 for Gradient Boosting. This confirms that vein-pattern features and rule-based scores provide meaningful insights in combination with other feature sets. Finally, the combination of five feature sets, color, vein, rule-based, spatial, texture resulted in the highest performance across the three classifiers.

Fig. 6 provides a summary of the classification accuracy obtained by the SVM classifier for all the feature combinations – single, two, three, four and the proposed five-feature set, reaching the highest accuracy of 0.82. This progressive improvement confirms that each feature domain – color, spatial, texture, vein-pattern and rule-based scores – contributes incrementally to the overall discriminative capacity of the framework, with no individual category able to emulate the performance of the full multi-domain representation. This analysis proves that integrating vein-pattern and rule-based features with color and texture information contributes to more reliable nutrient deficiency classification in the Ash gourd plant variety.

4.3. Comparison with existing feature extraction methods

The performance of three feature extraction methods, GLCM, HOG, and LBP, was evaluated with the classifiers Random Forest, SVM and Gradient Boosting over the Magnesium, Iron and Healthy classes, as shown in Table 7. Among the three methods, LBP had the most competitive overall performance compared to GLCM and HOG, with an accuracy ranging from 0.71 to 0.75. GLCM had accuracies of 0.69, 0.70, and 0.67 for Random Forest, SVM, and Gradient Boosting, respectively. HOG achieved the lowest average accuracy, with individual classifier accuracies ranging from 0.63 to 0.73.

Fig. 7 presents an accuracy comparison between the existing feature extraction and the proposed feature set for the Random Forest, SVM, and Gradient Boosting classifiers. The proposed framework with an SVM classifier outperforms traditional feature extraction and classifiers emphasizing the importance of a feature set comprising color, texture, spatial, vein, and rule-based features.

4.4. Comparison with existing works

As part of the overall study, the proposed method was compared with existing works, as shown in Table 8. Most existing studies relied on datasets with simple or fixed backgrounds, typically single-leaf images collected under greenhouse or standard controlled conditions, with feature sets largely limited to color, shape, geometry, and texture-domain descriptors such as DWT, GLCM, Gabor, LBP, and deep learning models. In contrast, the proposed framework incorporates domain-specific features-vein-pattern analysis and a rule-based scoring approach in addition to standard color, texture, and spatial features results in an efficient way to classify the nutrient deficiencies in the Ash gourd plant variety under in-field conditions.

5. Conclusion

The proposed AshGdNutDefAM framework explores the detection of magnesium and iron nutrient deficiencies in the Ash gourd plant variety by combining rule-based and vein-based features with handcrafted features. Among the three evaluated models, SVM demonstrated the highest accuracy, highlighting its potential to efficiently classify the Magnesium, Iron, and Healthy classes. This study highlights the importance of feature engineering in precision agriculture and timely identification; it also minimizes crop stress and increases crop yield. The

Table 6
Performance of four-feature combinations of color, spatial, texture, vein, rule-based features using random forest, SVM and gradient boosting classifiers.

Feature Set	Features	Random Forest				SVM				Gradient Boosting			
		Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
Spatial +Texture +Vein +Rule	35	0.71	0.73	0.71	0.72	0.69	0.68	0.69	0.67	0.66	0.71	0.68	0.70
Color +Texture +Vein +Rule	40	0.76	0.76	0.75	0.74	0.75	0.74	0.76	0.73	0.67	0.68	0.67	0.66
Color +Spatial +Vein +Rule	46	0.78	0.78	0.75	0.76	0.79	0.79	0.77	0.79	0.71	0.72	0.71	0.70
Color +Texture +Spatial+Rule	47	0.77	0.79	0.77	0.78	0.78	0.79	0.78	0.78	0.69	0.69	0.67	0.69
Color +Texture +Spatial +Vein	52	0.75	0.76	0.74	0.73	0.77	0.78	0.77	0.76	0.76	0.75	0.77	0.75

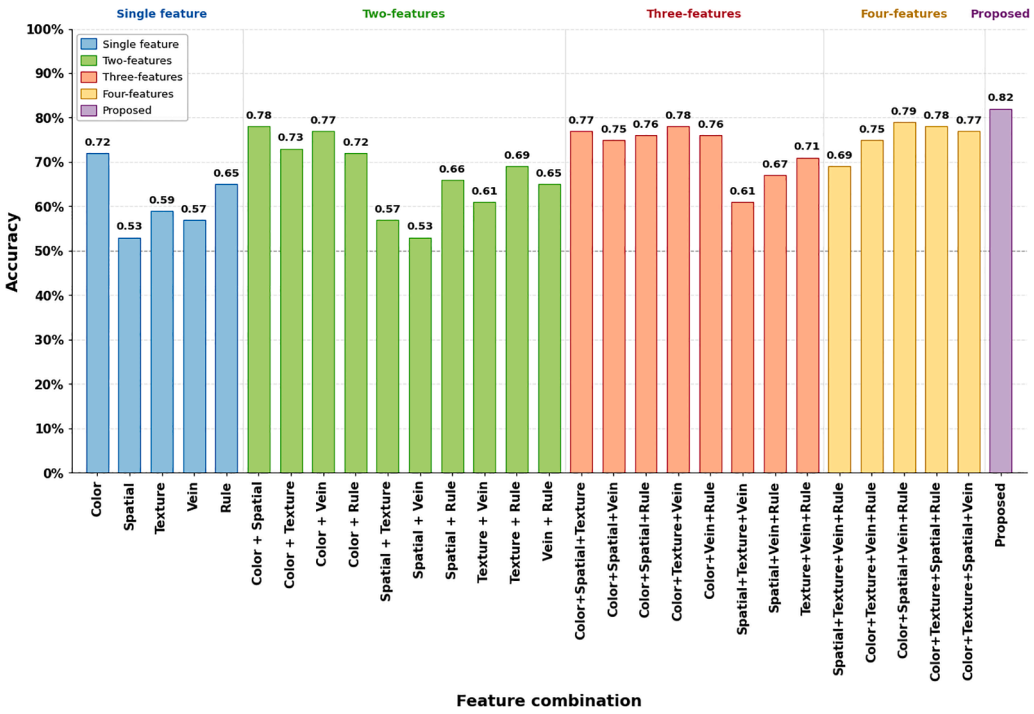


Fig. 6. Classification results using an individual feature set, two features, three features, four features, and the proposed feature set with the SVM classifier, showing the incremental improvement in classification performance obtained using multiple handcrafted feature sets.

Table 7
Performance of existing feature extraction methods-GLCM, LBP and HOG methods against Random Forest, SVM, and Gradient Boosting classifiers for Healthy, Magnesium and Iron classes.

		GLCM				LBP				HOG			
Classifier	Class	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
Random Forest	Healthy	0.70	0.71	0.41	0.52	0.75	0.80	0.79	0.80	0.67	0.42	0.25	0.31
	Magnesium	0.68	0.68	0.75	0.71	0.74	0.79	0.65	0.72	0.66	0.81	0.90	0.79
	Iron	0.69	0.75	0.88	0.81	0.75	0.66	0.82	0.73	0.67	0.60	0.70	0.64
SVM	Healthy	0.72	0.70	0.58	0.63	0.74	0.73	0.91	0.81	0.73	0.58	0.58	0.58
	Magnesium	0.70	0.66	0.70	0.68	0.76	0.84	0.80	0.82	0.71	0.81	0.89	0.84
	Iron	0.69	0.77	0.82	0.80	0.75	0.73	0.64	0.68	0.70	0.73	0.64	0.68
Gradient Boosting	Healthy	0.68	0.62	0.41	0.50	0.71	0.80	0.66	0.72	0.65	0.50	0.33	0.40
	Magnesium	0.67	0.73	0.70	0.71	0.70	0.88	0.75	0.81	0.64	0.76	0.80	0.78
	Iron	0.68	0.72	0.94	0.82	0.71	0.63	0.82	0.71	0.63	0.60	0.70	0.64

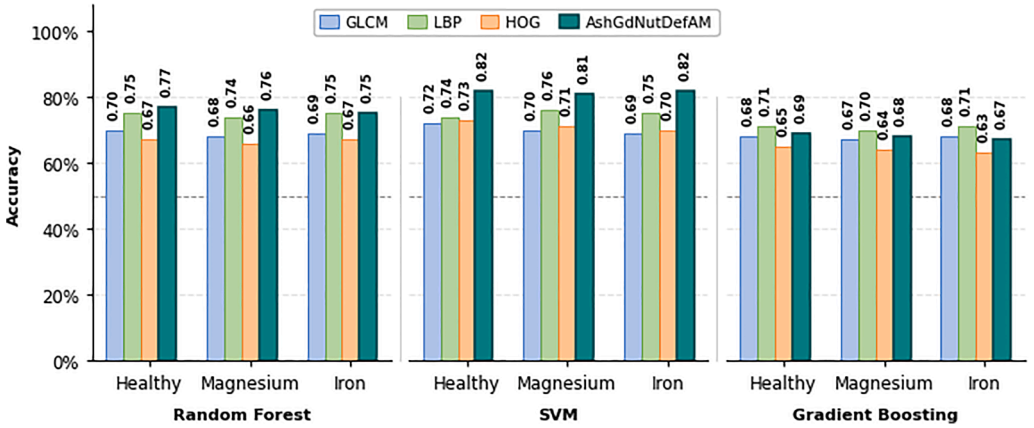


Fig. 7. Accuracy metric comparison of traditional methods GLCM, LBP, HOG, and the proposed AshGdNutDefAM method against Random Forest, SVM and Gradient Boosting classifiers for Healthy, Magnesium, and Iron classes.

dataset utilized for this study was collected from agricultural fields that contain image samples with complex backgrounds, varying lighting,

Table 8
Comparison of the proposed feature set, AshGdNutDefAM, with existing studies for plant classification and nutrient deficiency detection.

Ref.	Image characteristics	Features	Accuracy	Precision	Recall	F1
[23]	Single leaf and fixed plain background	Color, Texture, Shape	0.83	0.83	0.81	NA
[24]	Plain background single leaf	Color, Texture, Geometry	0.98	0.96	0.95	NA
[27]	Plant images from the pot and the field	DWT, GLCM, Gabor, LBP	0.67	0.71	0.67	NA
[4]	Plain background with images from the greenhouse	Color, morphological, and texture	0.90	0.90	0.88	0.87
[31]	Controlled, standard background	Color, Texture	0.81	NA	NA	NA
[28]	Single leaf image with standard plain background	Adam optimizer with the squeezeNet model	0.97	0.91	0.91	0.91
[29]	Single leaf image with plain background	Color	0.90	NA	NA	0.90
Proposed Method	Multiple leaves with a complex background	Color, Texture, Spatial, Vein, Rule-based	0.82	0.79	0.80	0.79

occlusion and shadows. Future research may utilize the multi-environment datasets and different creeper varieties to evaluate the performance, generalization, and stability of the model towards practical applications. Also, predicting deficiency severity and detecting multi-nutrient deficiencies in a single plant and leaf would enable continuous condition monitoring and timely targeted nutrient application.

6. Limitations

The dataset collected for this study was collected from a specific location in Mysuru, Karnataka, India, which leads to geographic and temporal bias. The visual symptoms caused by other stress factors such as water, heat stress, pest damage and disease may overlap with the symptoms caused by nutrient deficiency, leading to ambiguity at classification boundaries. Further, deficiency symptoms may also vary with crop age; solving this problem would require stage-wise classification models trained specifically for the plantation, mid-growth and late-growth stages. In some cases, plants may also suffer from multiple deficiencies simultaneously. The model requires validation on other cucurbits, such as pumpkin, gourd, and cucumber, to improve its generalizability across crop species. Further study on expected nutrient levels, alongside a visual comparison of deficiency severity, would enable continuous condition monitoring and timely, targeted nutrient application. Finally, the implementation of this model into large-scale agricultural systems integrating IoT and computer vision technologies for accurate nutrient monitoring is to be evaluated.

CRediT authorship contribution statement

M.A. Keerthi Prasad: Writing – review & editing, Validation, Formal analysis. **Pushpa B R:** Writing – review & editing, Validation, Supervision, Conceptualization. **Ardashir Mohammadzadeh:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] K.S. Parikh, P.P. Ghosh, H.P. Binswanger-Mkhize, Rapid economic growth in India: technical change in agriculture, irrigation and food security, Indian Econ. J. 64 (1–4) (2016) 115–136.
[2] C. Chatterjee, B.K. Dube, Nutrient deficiency disorders in vegetables and their management. Fruit and Vegetable Diseases, Springer Netherlands, Dordrecht, 2004, pp. 145–188.
[3] V.V. Carmona, L.C. Costa, Symptoms of nutrient deficiencies on cucumbers, Int. J. Plant Soil Sci. 8 (6) (2015) 1–11.
[4] M. Nadafzadeh, A. Banakar, S.A. Mehdizadeh, S. Minaei, A.M. Mouazen, G. Hoogenboom, Identification of nutrient deficiency stress for iron, zinc and manganese in baby spinach using computer vision, Smart Agricult. Technol. 12 (2025) 101524.
[5] A.N. Ganeshamurthy, H.B. Raghu-pathi, T.R. Rupa, S. Rajendiran, D. Kalaivanan, Micronutrient management in horticultural crops, Indian J. Fertilis. 14 (4) (2018) 68–85.
[6] P. Gupta, S. Chikkala, P. Kundu, Ash gourd and its applications in the food, pharmacological and biomedical industries, Int. J. Veg. Sci. 27 (1) (2021) 44–53.
[7] C.N.S. Campos, G.C.M. Teixeira, R.D.M. Prado, G. Caione, G.B. da Silva Júnior, C. H.O.D. David, P.E. Teodoro, Macronutrient deficiency in cucumber plants: impacts in nutrition, growth and symptoms, J. Plant Nutr. 44 (17) (2021) 2609–2626.
[8] A. Borlotti, G. Vigani, G. Zocchi, Iron deficiency affects nitrogen metabolism in cucumber (Cucumis sativus L.) plants, BMC Plant Biol. 12 (1) (2012) 189.
[9] B. Zhang, I. Cakmak, J. Feng, C. Yu, X. Chen, D. Xie, Y. He, Magnesium deficiency reduced the yield and seed germination in wax gourd by affecting the carbohydrate translocation, Front. Plant Sci. 11 (2020) 797.
[10] N. Vullaganti, B.G. Ram, X. Sun, Precision agriculture technologies for soil site-specific nutrient management: a comprehensive review, Artif. Intell. Agricult. 15 (2) (2025) 147–161.
[11] L. Sieczko, K. Kowalczyk, J. Gajc-Wolska, W. Kowalczyk, P. Dąbrowski, W. Borucki, H.M. Kalaji, Phosphorus-deficiency stress in cucumber (Cucumis sativus L.) plants: early detection based on chosen physiological parameters and statistical analyses, Photosynthetica 62 (1) (2024) 44–57.
[12] A.K. Ghorai, S. Mukhopadhyay, S. Kundu, S.N. Mandal, A.R. Barman, M. De Roy, S. Dutta, Image processing-based detection of diseases and nutrient deficiencies in plants, SATSA Mukhapatra 25 (1) (2021) 1–24.
[13] C. Dimkpa, P. Bindraban, J.E. McLean, L. Gatere, U. Singh, D. Hellums, Methods for rapid testing of plant and soil nutrients. Sustainable Agriculture Reviews, Springer International Publishing, Cham, 2017, pp. 1–43.
[14] M. Sudhakar, R.S. Priya, Computer vision based machine learning and deep learning approaches for identification of nutrient deficiency in crops: a survey, Nat. Environ. Pollut. Technol. 22 (3) (2023) 1387–1399.
[15] J. Yu, J. Wang, B. Leblon, Y. Song, Nitrogen estimation for wheat using UAV-based and satellite multispectral imagery, topographic metrics, leaf area index, plant height, soil moisture, and machine learning methods, Nitrogen 3 (1) (2021) 1–25.
[16] N. Paul, G.C. Sunil, D. Horvath, X. Sun, Deep learning for plant stress detection: a comprehensive review of technologies, challenges, and future directions, Comput. Electron. Agricult. 229 (2025) 109734.
[17] S. Sharma, M. Vardhan, MTJNet: multi-task joint learning network for advancing medicinal plant and leaf classification, Knowl. Based Syst. 299 (2024) 112147.
[18] O. Ennaji, L. Vergütz, A. El Allali, Machine learning in nutrient management: a review, Artif. Intell. Agricult. 9 (2023) 1–11.
[19] V. Aleksandrov, Identification of nutrient deficiency in plants by artificial intelligence, Acta Physiol. Plant. 44 (3) (2022) 29.

[20] J. Shi, Y. Wang, Z. Li, X. Huang, T. Shen, X. Zou, Simultaneous and nondestructive diagnostics of nitrogen/magnesium/potassium-deficient cucumber leaf based on chlorophyll density distribution features, Biosyst. Eng. 212 (2021) 458–467.
[21] Q. Peng, W. Xu, Crop nutrition and computer vision technology, Wirel. Personal Commun. 117 (2) (2021) 887–899.
[22] J.G.A Barbedo, Detection of nutrition deficiencies in plants using proximal images and machine learning: a review, Comput. Electron. Agricult. 162 (2019) 482–492.
[23] D. Rahadiyan, S. Hartati, A.P. Nugroho, Feature aggregation for nutrient deficiency identification in chili based on machine learning, Artif. Intell. Agricult. 8 (2023) 77–90.
[24] S. Farzizadeh, H.N. Mardanqom, Smart detection of plant nutrient deficiencies using machine learning and image fusion, APL Mach. Learn. 3 (4) (2025).
[25] R.G. Pinheiro, J.G. Lopes, M.M. Souza, F.N. Medeiros, Plant leaf classification using the multiscale entropy of curvature and feature aggregation, Ecol. Inform. 91 (2025) 103373.
[26] D.M.G.D. Rathnayake, K.M.S.J. Kumarasinghe, R.M.I.K. Rajapaksha, N.K.A. C. Katuwawala, Green Insight: a novel approach to detecting and classifying macro nutrient deficiencies in paddy leaves, in: 2023 8th International Conference on Information Technology Research (ICITR), IEEE, 2023, pp. 1–6.
[27] A. Patel, R. Mishra, A. Sharma, Classification nutrient deficiency of maize plant leaf using machine learning algorithm, Fusion Pract. Appl. 17 (2025) 78–94.
[28] F.D. Shadrach, G. Kandasamy, S. Neelakandan, T.B. Lingaiah, Optimal transfer learning based nutrient deficiency classification model in ridge gourd (Luffa acutangula), Sci. Rep. 13 (1) (2023) 14108.
[29] N.A. Orka, E. Haque, M.N. Uddin, T. Ahamed, Nutrispace: a novel color space to enhance deep learning based early detection of cucurbits nutritional deficiency, Comput. Electron. Agricult. 225 (2024) 109296.
[30] S. Sakthipriya, R. Naresh, Precision agriculture based on convolutional neural network in rice production nutrient management using machine learning genetic algorithm, Eng. Appl. Artif. Intell. 130 (2024) 107682.
[31] M.S. Sirsat, D. Isla-Cernadas, E. Cernadas, M. Fernández-Delgado, Machine and deep learning for the prediction of nutrient deficiency in wheat leaf images, Knowl. Based Syst. 317 (2025) 113400.
[32] D. Rahadiyan, S. Hartati, A.P. Nugroho, Design of an intelligent hydroponics system to identify macronutrient deficiencies in chili, Int. J. Adv. Comput. Sci. Appl.s 13 (1) (2022).
[33] A. Bera, D. Bhattacharjee, O. Krejcar, PND-Net: plant nutrition deficiency and disease classification using graph convolutional network, Sci. Rep. 14 (1) (2024) 15537.
[34] R. Sathyavani, K. JaganMohan, B. Kalaavathi, Classification of nutrient deficiencies in rice crop using denseNet-BC, Mater. Today: Proc. 56 (2022) 1783–1789.
[35] M.S. Sirsat, D. Isla-Cernadas, E. Cernadas, M. Fernández-Delgado, Machine and deep learning for the prediction of nutrient deficiency in wheat leaf images, Knowl. Based Syst. 317 (2025) 113400.